

Module 1-CSS3

Exercise 1

Problem Statement:

Why CSS? Inline vs. Internal vs. External

Scenario: The designer wants you to experiment with different ways to apply styles.

Objective: Understand various CSS inclusion methods and their impact.

Task:

- Apply an inline style to make one heading red.
- Use an embedded <style> tag in the <head> to define body background.
- Link an external stylesheet styles.css and move all reusable styles there.
- Add comments in your CSS to label each section (/* Header styles */)

Code:

E1.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Exercise 1</title>

  <!-- Internal CSS -->
  <style>
    body{
      background-color:#f0f8ff;
    }
  </style>

  <!-- External CSS -->
  <link rel="stylesheet" href="styles.css">
</head>
<body>

  <h1 style="color:red;">Local Community Event Portal</h1>

  <div class="eventCard">
    Welcome to Community Events
  </div>
```

```
</body>
```

```
</html>
```

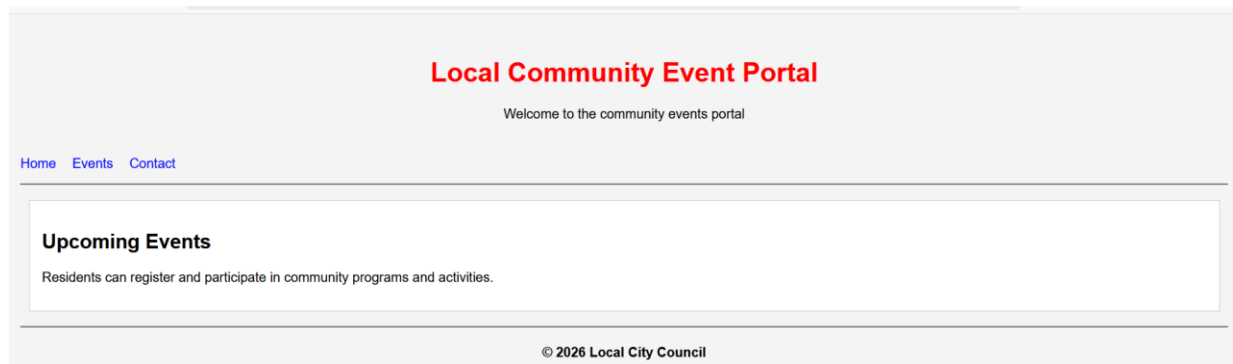
E1.css

```
/* Event Card Styles */
```

```
.eventCard{
  border:1px solid black;
  padding:15px;
  background-color:white;
  width:250px;
```

}

Output:



Explanation:

- CSS can be applied using inline, internal, and external methods.
- External CSS improves reusability and maintenance.

Exercise 2

Problem Statement:

CSS Syntax and Comments

Scenario: You've joined a team and need to understand and maintain a large stylesheet.

Objective: Write clean, readable CSS with proper structure and comments.

Task:

- Create a section in styles.css with formatted rules and consistent indentation.
- Add descriptive comments above selectors.
- Example:

```
/* Style for main CTA button */
```

```
.cta-button {  
background-color: #007BFF;  
color: white;  
}
```

Code:

E2.html

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>Exercise 2</title>  
  
  <link rel="stylesheet" href="styles.css">  
</head>  
<body>  
  
  <button class="cta-button">  
    Register Now  
  </button>  
  
  <div class="eventCard">  
    Music Festival Registration Open  
  </div>  
  
</body>  
</html>
```

E2.css

```
/* Style for Main CTA Button */
```

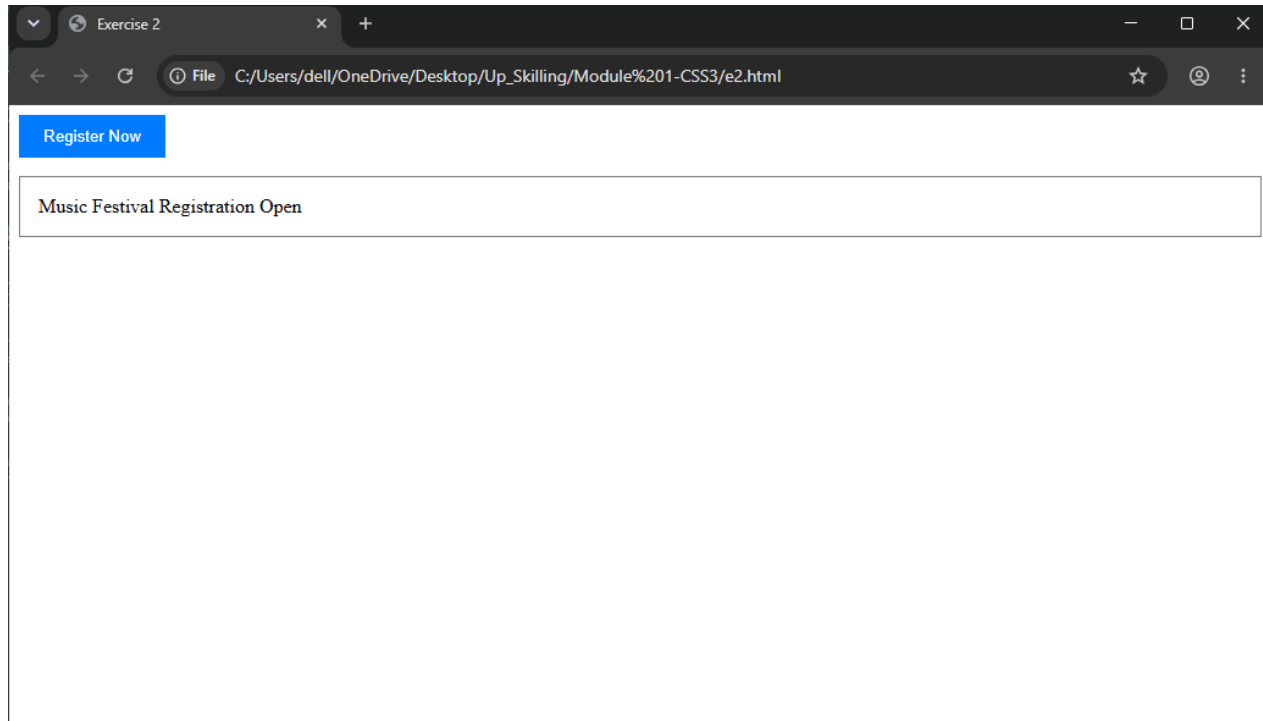
```
.cta-button {  
  background-color: #007BFF;  
  color: white;  
  padding: 10px 20px;  
  border: none;  
  cursor: pointer;  
}
```

```
/* Event Card Styling */
```

```
.eventCard {
```

```
border:1px solid gray;  
padding:15px;  
margin-top:15px;  
}
```

Output:



Explanation:

- CSS comments help developers understand stylesheet sections.
- Proper formatting makes CSS easier to maintain.

Exercise 3

Problem Statement:

Selectors Playground

Scenario: You need to style various elements based on IDs, classes, and element types.

Objective: Master different selector types.

Task:

- Use:
 - o Universal selector * to reset margin/padding
 - o Element selector to style all <h2>
 - o ID selector #mainHeader for the banner
 - o Class selector .eventCard for event containers
 - o Grouping selector for h3, p to style together

Code:

E3.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Exercise 3</title>

  <link rel="stylesheet" href="styles.css">
</head>
<body>

  <div id="mainHeader">
    Community Event Portal
  </div>

  <h2>Upcoming Events</h2>

  <div class="eventCard">
    Sports Meet 2026
  </div>

  <h3>Music Festival</h3>

  <p>Enjoy live performances.</p>

</body>
</html>
```

E3.css

```
/* Universal Selector */

*{
  margin:0;
  padding:0;
  box-sizing:border-box;
}

/* Element Selector */
```

```
h2{
  color:blue;
  margin:20px;
}

/* ID Selector */

#mainHeader{
  background-color:orange;
  padding:20px;
  font-size:24px;
}

/* Class Selector */

.eventCard{
  border:2px solid black;
  margin:20px;
  padding:15px;
}

/* Grouping Selector */

h3,p{
  color:green;
  margin:10px;
}
```

Output:

Local Community Event Portal

Music Concert

Community Gathering

Enjoy live music performances and cultural activities.

Sports Event

Annual Sports Meet

Participate in outdoor games and fitness activities.

Explanation:

- Different selectors target elements in different ways.
- Grouping selectors apply common styles to multiple elements.

Exercise 4

Problem Statement:

Color & Background Styling

Scenario: You're theming the portal based on a city council's branding.

Objective: Apply consistent colors and background visuals.

Task:

- Use HEX and RGBA for setting text and background colors
- Add a background image to the body with fallback color
- Apply gradients to section headers using background: linear-gradient(...)

Code:

E4.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Exercise 4</title>

  <link rel="stylesheet" href="styles.css">
</head>
<body>

  <h1>Local Community Event Portal</h1>

  <h2 class="sectionHeader">
    Upcoming Events
  </h2>

  <div class="eventCard">
    Annual Food Festival
  </div>

</body>
</html>
```

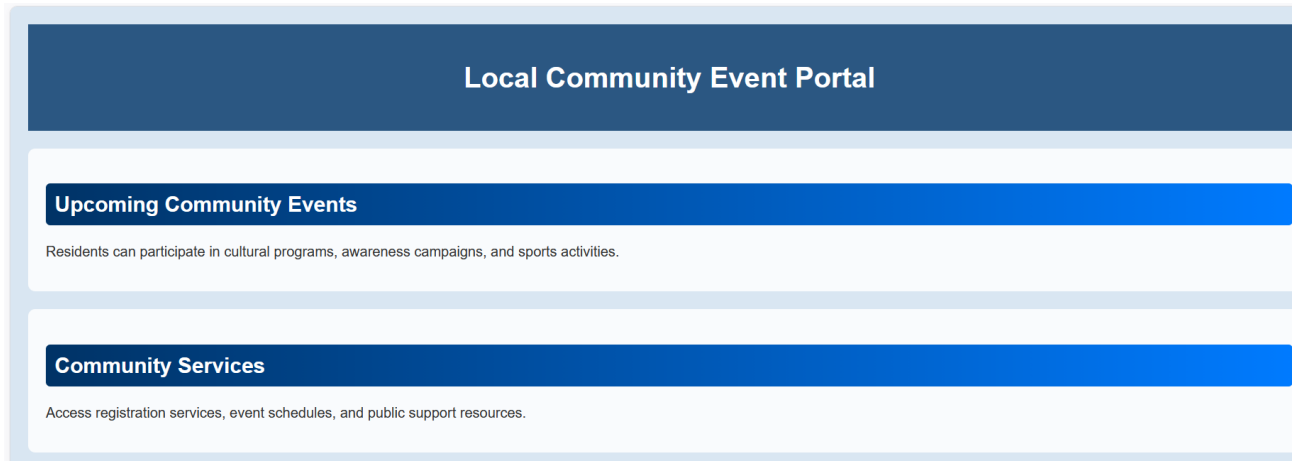
E4.cs

```
s
body{
  background-color:#f5f5f5;
  background-image:url("background.jpg");
  background-size:cover;
}

h1{
  color:#003366;
}

.eventCard{
  background-color:rgba(0,123,255,0.3);
  padding:20px;
  margin:20px;
}
```

```
.sectionHeader{
  background:linear-gradient(to right, blue, lightblue);
  color:white;
  padding:15px;
}
```



Explanation:

- HEX and RGBA colors provide flexible color styling.
- Gradients and background images improve visual appearance.

Exercise 5

Problem Statement:

Typography: Fonts and Text

Scenario: The marketing team wants more appealing fonts and better readability. Objective: Enhance textual appearance using CSS properties.

Task:

- Use @import or <link> to include a Google Font
- Set font-family, font-size, font-style, font-weight in different sections
- Use text-align, text-transform, letter-spacing, line-height on descriptions

Code:

E5.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Exercise 5</title>

  <link href="https://fonts.googleapis.com/css2?family=Roboto&display=swap" rel="stylesheet">

  <link rel="stylesheet" href="styles.css">
</head>
<body>

  <h1>Community Events</h1>

  <h2>Upcoming Activities</h2>

  <p class="description">
    Community members can participate in exciting local events and social gatherings.
  </p>

</body>
</html>
```

E5.css

```
body{
  font-family:'Roboto',sans-serif;
}

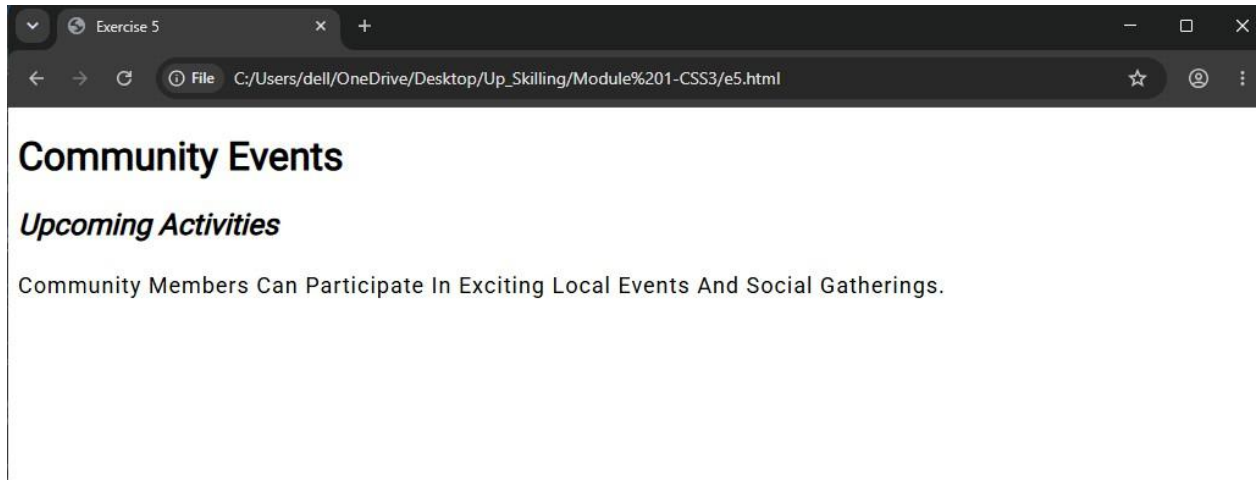
h1{
  font-size:32px;
  font-weight:bold;
}

h2{
  font-style:italic;
}

.description{
  text-align:justify;
  text-transform:capitalize;
```

```
letter-spacing: 1px;  
line-height: 1.8;  
font-size: 18px;  
}
```

Output:



Explanation:

- Typography properties improve readability and presentation.
- Google Fonts provide modern and attractive font styles.